

Notes on sampling

What we want to be able to do, is to obtain an effects matrix B of the confounders on variables X and Y . Now these effect sizes should be chosen in a very particular way, to reflect how strongly the confounders affect the variables as a whole, and how much of that effect is caused by a non-linear component of the confounders.

Defining the parameters

We take a number of base confounders, which exert a linear effect on the variables X and Y . We call this number k_{lin} . Now every confounder also has a number of non-linear components, which exert a non-linear effect on the variables X and Y . We call the sum over the amount of these variables for all confounders k_{nonlin} . The total number of confounders is therefore $k = k_{\text{lin}} + k_{\text{nonlin}}$. In the derivations below we will first work this out for the case where each confounder has only one non-linear component. Later on we will extend this to multiple non-linear components.

Now the k confounders exert an effect on the variables X and Y . These effects can be captured in the vectors:

$$\gamma_x \in \mathbb{R}^k$$

and

$$\gamma_y \in \mathbb{R}^k$$

meaning the entries to both vectors can take on any real number. Both vectors together can be written as

$$\theta = (\gamma_x, \gamma_y) \in \mathbb{R}^{2k}.$$

Now we constrain each coefficient to be between 0.01 and 0.30, since larger values seem unrealistic and may lead to difficulties to get the system stationary. However, we could always opt for a larger upper bound if needed. We can summarise this with the following expression:

$$\Theta = \{ \theta = (\gamma_x, \gamma_y) : \gamma_x, \gamma_y \in [0.01, 0.30]^k \}.$$

The influence of confounders

We can compute the total confounder induced variance in X and Y as follows:

$$S_U = \begin{pmatrix} \text{Var}_X(U) & \text{Cov}_{XY}(U) \\ \text{Cov}_{XY}(U) & \text{Var}_Y(U) \end{pmatrix} = \begin{pmatrix} \sum_j \gamma_{xj}^2 & \sum_j \gamma_{xj} \gamma_{yj} \\ \sum_j \gamma_{xj} \gamma_{yj} & \sum_j \gamma_{yj}^2 \end{pmatrix}.$$

Now when the confounders are standard normal and uncorrelated, we can simplify this to:

$$\text{Var}_X^U = \sum_{i=1}^k \gamma_{x,i}^2.$$

Adding any covariances just complicates this calculation a little bit, but is not all that difficult. Now we can write the total influence of the confounders as:

$$S(\theta) = \sum_{i=1}^k \gamma_{x,i}^2 + \sum_{i=1}^k \gamma_{y,i}^2 = s_0.$$

Where later on, we want to pick s_0 .

Linear vs non-linear influence

Now we want to compute which part of this variance is attributable to a linear component, and which part is attributable to a non-linear component. This is a measure of non-linearity which falls between 0 and 1. We can partition our vectors of coefficients as follows:

$$\gamma_x = (\gamma_{x,\text{lin}}, \gamma_{x,\text{nl}}) \quad \gamma_y = (\gamma_{y,\text{lin}}, \gamma_{y,\text{nl}})$$

where the first k_{lin} entries are the linear parts, and the last k_{nonlin} entries are the non-linear parts. Now we can compute for both X and Y the non-linear part of the explained variance as:

$$\eta_X(\theta) = \sum_{i=k_{\text{lin}}+1}^k \gamma_{x,i}^2 \quad \eta_Y(\theta) = \sum_{i=k_{\text{lin}}+1}^k \gamma_{y,i}^2.$$

Then their average gives us the non-linear influence as:

$$\eta(\theta) = \frac{\eta_X + \eta_Y}{2} = \eta_0.$$

Where later we want to pick η_0 .

Sampling procedure

Now we can choose η_0 and s_0 . We can also choose a number of linear confounders k_{lin} and a number of non-linear components k_{nonlin} . We can call the number of non-linear components per confounder the complexity of the nonlinearity. As touched upon before, for now we will assume this complexity is 1, meaning each confounder has exactly one non-linear component. Now there is a set of parameters θ which satisfy these constraints. We can call the set of solutions to these constraints:

$$M = \{ \theta \in \Theta : S(\theta) = s_0, \eta(\theta) = \eta_0 \}.$$

Now the problem is, we do not know M , and as such we cannot sample from it. We can however sample from the set of all possible values in our $2k$ -dimensional

space Θ . Now we do something called rejection sampling, where we sample from this space and check if the constraints are met. If not, we reject the sample and sample again. Now, you might already see that this process will be incredibly inefficient, as picking a random sample from Θ will almost never satisfy the constraints.

There are some things we can do to make this process more efficient. First of all, we can make the constraints a little bit less strict, by allowing a small margin of error around the desired values. This will “thicken” the set M a little bit, making it more likely to hit a valid sample. Now, we also need to make sure that we pick our samples in a smarter way. In practice this means choosing one more constraint. We will choose to divide up s_0 evenly over X and Y . This means we add the constraint:

$$\text{Var}_X^U = \text{Var}_Y^U = \frac{s_0}{2}.$$

Note that we could always choose a different way to divide up s_0 over X and Y . We could each time also sample this division from a uniform distribution before picking any values from the space of the parameters. In this way we would generalise over different ways to divide up the confounder influence over X and Y . Note that we face a similar choice when dividing up η_0 over X and Y . For now we will keep it simple and divide both evenly:

$$\eta_X = \eta_Y = \eta_0.$$

Now recall the equation for η_X :

$$\eta_X = \frac{N_X}{L_X + N_X},$$

which we can now rewrite as:

$$N_X = \eta_0(L_X + N_X).$$

Now note that the total amount of variance $L_X + N_X$ is equal to $s_0/2$. Therefore we can rewrite this as:

$$N_X = \eta_0 \frac{s_0}{2},$$

and

$$N_Y = \eta_0 \frac{s_0}{2},$$

and

$$L_X = (1 - \eta_0) \frac{s_0}{2},$$

and

$$L_Y = (1 - \eta_0) \frac{s_0}{2}.$$

Now for each of these four quantities, we can calculate them using the sum of squares:

$$L_X = \sum_{i=1}^{k_{\text{lin}}} \gamma_{x,\text{lin},i}^2,$$

$$L_Y = \sum_{i=1}^{k_{\text{lin}}} \gamma_{y,\text{lin},i}^2,$$

$$N_X = \sum_{i=1}^{k_{\text{nonlin}}} \gamma_{x,\text{nl},i}^2,$$

$$N_Y = \sum_{i=1}^{k_{\text{nonlin}}} \gamma_{y,\text{nl},i}^2.$$

Now this is where we can do something smart. We want to draw a random vector from any symmetric distribution, such that the sum of squares of the entries:

$$u^{x,\text{lin}} \in \mathbb{R}^{k_{\text{lin}}},$$

we normalize this vector such that:

$$\sum_{i=1}^{k_{\text{lin}}} (u_i^{x,\text{lin}})^2 = 1,$$

and then we set:

$$\gamma_{x,\text{lin},i} = \sqrt{L_X} u_i^{x,\text{lin}},$$

which fulfills the constraint $L_X = \sum_{i=1}^{k_{\text{lin}}} \gamma_{x,\text{lin},i}^2$.

We can do the same for the non-linear confounders:

$$u^{x,\text{nl}} \in \mathbb{R}^{k_{\text{nonlin}}},$$

normalize:

$$\sum_{i=1}^{k_{\text{nonlin}}} (u_i^{x,\text{nl}})^2 = 1,$$

and set:

$$\gamma_{x,\text{nl},i} = \sqrt{N_X} u_i^{x,\text{nl}},$$

which fulfills the constraint $N_X = \sum_{i=1}^{k_{\text{nonlin}}} \gamma_{x,\text{nl},i}^2$.

We can do the same for Y . Now we also wanted our samples to fall within the range between 0.01 and 0.30. We can simply check this after sampling and reject any samples that do not fall within this range.

To recap in words, we:

- Choose k_{lin} , k_{nonlin} , s_0 and η_0
- Sample four random vectors from a symmetric distribution: $u^{x,\text{lin}}$, $u^{x,\text{nl}}$, $u^{y,\text{lin}}$, $u^{y,\text{nl}}$
- Normalize these vectors to have sum of squares equal to 1
- Scale these vectors to have sum of squares equal to their respective budgets
- Combine these vectors to form γ_x and γ_y
- Check if all entries of γ_x and γ_y fall within $[0.01, 0.30]$. If not, reject and start over.